



Answers & Solutions:

1. **A. Doubled**
2. **A. Absolute potential**
3. **D. It allows independent operations of loads**
4. **D. Both A and B.**
5. **C. 625 W**

Solution:

First obtain equivalent Thevenin circuit across load R_L

Thevenin voltage

$$\frac{V_{th} - 110\angle 0^\circ}{6 + j8} + \frac{V_{th} - 90\angle 0^\circ}{6 + j8} = 0$$

$$2V_{th} - 200\angle 0^\circ = 0$$

$$V_{th} = 100\angle 0^\circ \text{ V}$$

Thevenin impedance

$$Z_{th} = (6 + j8)\Omega \parallel (6 + j8)\Omega$$

$$Z_{th} = (3 + j4)\Omega$$

For maximum power transfer

$$R_L = |Z_{th}| = \sqrt{3^2 + 4^2} = 5 \Omega$$

Power in load

$$P = (i_{eff})^2 R_L$$

$$P = \left(\frac{100}{3 + j4 + 5} \right)^2 (5) = \frac{100^2}{80} (5)$$

$$P = 625 \text{ W} \rightarrow \text{Ans}$$

6. **A. 10**
7. **C. increases.**
8. **B. $R/3$**
9. **A. 69 Ω**

Solution:

$$R = R_o (1 + \alpha \Delta T)$$

$$R = 50 \Omega \left[1 + 0.0038\%_C (100 - 0) \right]$$

$$R = 69 \Omega \rightarrow \text{Ans.}$$

10. **B. 222.5 V**

Solution:

$$R_L = \rho \frac{L}{A}$$

$$R_L = 1.78 \mu\Omega - \text{cm} \frac{73.2 \text{ m} \times \frac{100 \text{ cm}}{1 \text{ m}}}{1.29 \text{ cm}^2}$$

$$V_L = IR_L = 250(0.01) = 2.5$$

$$V_s = V_L + V_o = 2.5 \text{ V} + 220 \text{ V}$$

$$V_s = 222.5 \text{ V} \rightarrow \text{Ans.}$$

11. **D. more heat**
12. **D. Double the resistance.**
13. **A. 64.81 Ω**

Solution:

$$\alpha_o = \frac{1}{|T| + 0}; |T| = 250$$

$$\alpha_{20} = \frac{1}{250 + 20} = 0.0037\%_C$$

$$R = R_{20} (1 + \alpha_{20} \Delta T)$$

$$R = 50 \Omega \left[1 + 0.0037\%_C (100 - 20) \right]$$

$$R = 64.81 \Omega \rightarrow \text{Ans.}$$

14. **B. shorting the load terminals**
15. **C. three-terminal**
16. **D. four times**
17. **A. Two**
18. **A. 10 ohms**

Solution:

$$R_T = \{ 20 \parallel [(20 \parallel 20) + 10] \}$$

$$R_T = [20 \parallel (10 + 10)]$$

$$R_T = 10 \Omega \rightarrow \text{Ans.}$$

19. **A. Four times as great**
20. **B. the same everywhere.**
21. **B. zero**

22. **C. Conservation of energy**

23. **A. 0.52 A**

Solution:

$$I_T = \frac{230 \text{ V}}{R_T}$$

$$I_T = \frac{230 \text{ V}}{R_A + R_B}$$

$$I_T = \frac{230 \text{ V}}{176.33 \Omega + 264.5 \Omega}$$

$$I_T = 0.52 \text{ A} \rightarrow \text{Ans.}$$

where:

$$R_T = R_A + R_B = 440.83 \Omega$$

$$R_A = \frac{V_A^2}{P_A} = \frac{(115)^2}{75} = 176.33 \Omega$$

$$R_B = \frac{V_B^2}{P_B} = \frac{(115)^2}{50} = 264.5 \Omega$$

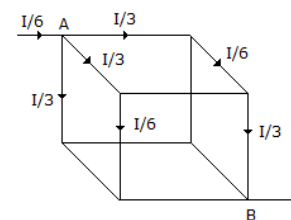
24. **D. no change**
25. **A. Ohm's law**
26. **B. Nichrome**
27. **C. 0 V.**
28. **D. Voltage divider**
29. **B. Thevenin's theorem**
30. **D. Both A and C.**

31. **A. 5/6 Ω**

Solution:

$$V_{AB} = IR = \left(\frac{1}{6} + \frac{1}{3} + \frac{1}{3} \right) (1) = \frac{5I}{6}$$

$$\therefore \frac{V_{AB}}{I} = \frac{5}{6} \Omega \rightarrow \text{Ans.}$$





32. **A. the direction of current through the resistor.**

33. **B. 1333**

Solution:

$$W_{in} = W_{out} + W_{loss}$$

$$W_{in} = 1000 + 0.25W_{in}$$

$$W_{in} = 1333.333 \text{ kWh} \rightarrow \text{Ans.}$$

34. **A. 960 Ω**

Solution:

$$\frac{R_1}{L_1} = \frac{R_2}{L_2}$$

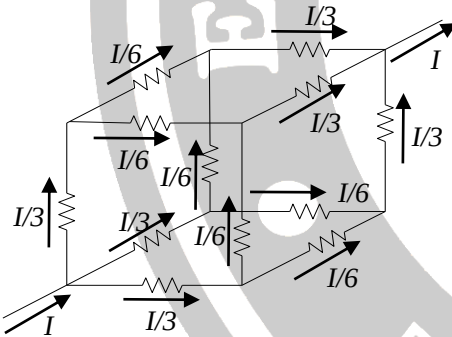
$$\frac{600 \Omega}{5 \text{ m}} = \frac{R_2}{8 \text{ m}}$$

$$R_2 = 960 \Omega \rightarrow \text{Ans.}$$

35. **C. Provides more power**

36. **A. 0.20 V**

Solution:



Let R = resistance of each resistor

$$R_t = \frac{5}{6}R$$

$$R_t = \frac{5}{6}(6)$$

$$R_t = 5 \Omega$$

$$V_t = I_t R_t$$

$$V_t = (40 \times 10^{-3})(5)$$

$$V_t = 0.20 \text{ V} \rightarrow \text{Ans.}$$

37. **C. number of voltage sources**

38. **A. independent of each other.**

39. **D. Loop**

40. **C. Provides more power**

41. **C. the remaining branch currents do not change in value.**

42. **C. inversely proportional to the branch resistance values.**

43. **C. Superposition theorem**

44. **B. 18 A and 12A**

Solution:

$$I_T = I_{R_1} + I_{R_2}$$

$$I_{R_1} = I_T \frac{4}{4+6} = 30 \frac{4}{10} = 12 \text{ A} \rightarrow \text{Ans.}$$

$$I_{R_2} = 30 - 12 = 18 \text{ A} \rightarrow \text{Ans.}$$

45. **C. Reciprocity theorem**

46. **C. The 100- Ω resistor.**

47. **C. equal to**

48. **A. 1507965 sq. mil**

Solution:

$$1 \text{ mil} = 0.001 \text{ in}$$

$$A = \pi r^2 (12) = \pi \left(0.2 \text{ in} \times \frac{1 \text{ mil}}{0.001 \text{ in}} \right)^2 (12)$$

$$A = 1507965 \text{ sq.mil} \rightarrow \text{Ans.}$$

49. **B. the 10- Ω resistor.**

50. **B. open the load under test**

END